Roll Number: _____

Thapar Institute of Engineering and Technology
Department of Computer Science and Engineering

B.E. COE: Semester-VII Auxiliary Examination **Course Code: UCS 802**
Course Name: Compiler Construction

February 23, 2025, at 10.00 AM **Name of Faculty: Geeta Kasana**
Time: 3 Hours M. Marks: 100

Note: Attempt to answer all questions with proper ***justification***. Assume missing data, if any, suitably.

Q1	Convert the regular expression $r = (0+1) * 01(0 + 1))$ into NFA using Thompson's construction and convert the obtained into DFA using the subset construction method	10
Q2	Discuss all four variants of bottom-up parsers with the basic differences.	20
Q3	a) Construct the Syntax Tree and Directed Acyclic Graph (DAG) for the following expression: $(x * y + z)/(p - q) + r * (p - q)$ b) Consider the following statement $x = y + z * 3$ Explain each phase and show all phases pictorially which a compiler will follow in compiling the above statement.	20
Q4	a) Explain how error recovery techniques differ between top-down and bottom-up parsers. b) Discuss the differences between the ***Synthesized*** attribute and the ***Inherited*** attribute.	10
Q5	Consider the following expression: $(A - B * C)/(D + E) + (F * G - H)$ (a) Write the sequence of three-address instructions generated for the above expression. (b) Represent the Quadruples, Triples, and Indirect-Triple implementation for the above three-address code.	10
Q6	Consider the following grammar: $E \rightarrow E + T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow (E) \mid id$ a) Remove the left recursion from the above grammar. b) Construct First and Follow sets for the non-terminals of the resulting grammar. c) Construct LL(1) parsing table.	10
Q7	Differentiate between the following with suitable examples: a) Dead Code Elimination and Loop Optimization. b) Abstract Syntax Tree and Directed Acyclic Graphs. c) Constant Propagation and Constant Folding d) Stack and Heap based memory allocation	20

********** All The Best**********